

Data Quality Analysis Report

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Executive Summary

This report analyzes data quality across 1 files containing 43,449 total records. The analysis identified 197,499 validation errors. **Key Recommendations:** • Address mandatory field completeness issues at the source • Implement data validation workflows before processing • Review critical validation errors for immediate remediation

File Quality Scores

File Name	Mandatory Complete	Total Records	Errors	Quality %
Core_Employee.xlsx	100.0%	43,449	197,499	41.4%

Error Summary

Error Type	Count	Severity	Recommended Action
TYPE_COMPATIBILITY	86,898	Critical	Workshop: incompatible types. Resolve compatibility.
HETEROGENEOUS_TYPE	75,572	High	Workshop: mixed data types. Standardize approach needed.
MANDATORY_EMPTY	18,513	Critical	Source: mandatory field empty. Requires data correction.
TRUNCATION	15,128	High	Source: field exceeds length limit. Adjust target or truncate data.
ENCODING_ISSUE	1,388	High	Source: invalid characters. Fix character encoding.

Cross-File Referential Integrity Analysis

This section analyzes referential integrity between files using dynamic relationship discovery from Neo4j field mappings. Critical integrity issues can impact data migration success.

Activities Domain Integrity Analysis

Business rule-based orphan detection for Activities domain relationships. This analysis identifies ILT courses and sessions that lack valid activity references.

High Priority Issues (1 relationships)

Relationship	Key Field	Integrity %	Orphaned	Business Impact
Activities.ILTCourseCode → Activities.Activity Code	ILTCourseCode →Activity Code	0.0%	162.0	High: 162/None records with integrity issues - impacts data quality

Medium Priority Issues (4 relationships)

Relationship	Integrity %	Orphaned Records
Activities.ClassCode → Activities.Activity Code (report-only)	0.0%	451.0
Activities.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	451.0
Activities.ILTCourseCode → Activity Links.Child ActivityCode (report-only)	0.0%	157.0
Activities.SessionParts.ClassCode → Activity Links.Child ActivityCode (report-only)	0.0%	1.0

Employees Domain Integrity Analysis

Business rule-based orphan detection for Employee domain relationships. This analysis identifies transcript records, instructor assignments, and curriculum ownership that lack valid employee references.

High Priority Issues (6 relationships)

Relationship	Key Field	Integrity %	Orphaned	Business Impact
Prereq_Instr.EmployeeId → PersonNumber	EmployeeId→PersonNumber	0.0%	3.0	High: 3/None instructor assignments invalid - impacts training delivery authorization
Prereq_Instr.EmployeeId → PersonNumber	EmployeeId→PersonNumber	96.7%	3.0	High: 3/90 instructor assignments invalid - impacts training delivery authorization
Prereq_Instr.EmployeeId → PersonNumber	EmployeeId→PersonNumber	96.7%	3.0	High: 3/90 instructor assignments invalid - impacts training delivery authorization
Transcript.*.EmployeeID → PersonNumber	EmployeeID→PersonNumber	100.0%	0.0	High: 0/0 training records lack valid employee references - impacts completion tracking and compliance
Transcript.*.EmployeeID → PersonNumber	EmployeeID→PersonNumber	100.0%	0.0	High: 0/0 training records lack valid employee references - impacts completion tracking and compliance
Transcript.*.EmployeeID → PersonNumber	EmployeeID→PersonNumber	100.0%	0.0	High: 0/0 training records lack valid employee references - impacts completion tracking and compliance

Error generating cross-file analysis: '<' not supported between instances of 'NoneType' and 'int'. Check system configuration.

Cross-File Error Patterns (Legacy Analysis)

No cross-file error patterns detected.